

CPU Scheduling for a Bank Teller / ATM Transaction Server

Individual Contribution: Process Management and CPU Scheduling

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1. Scope and Parameters

This is my part of the group assignment. I modelled the server as five concurrent processes and compared four scheduling policies on waiting time, turnaround time and fairness. The processes are P1(0,3), P2(2,6), P3(4,2), P4(6,8), P5(8,4) as (arrival, burst), with a Round Robin quantum of 3. Priorities are P1=2, P2=3, P3=1, P4=4, P5=2, smaller meaning higher: P3 is an urgent authorisation check, P4 a long low-urgency reconciliation. Total burst is 23 units, so every policy ends at $t = 23$ and only the ordering differs.

2. Schedules Produced

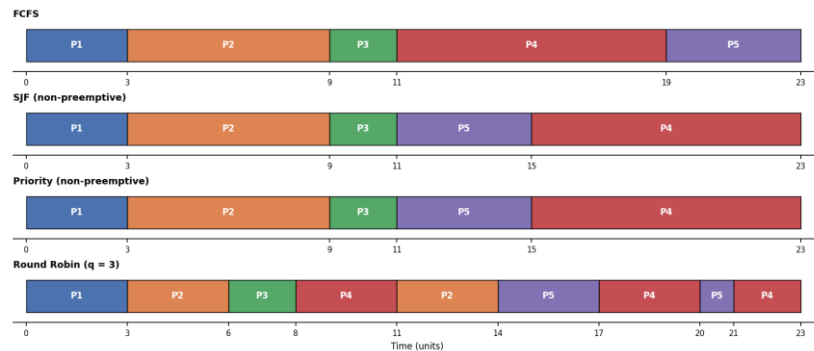


Figure 1. Gantt charts for the four policies on a common time axis.

3. Results

Policy	Execution order	Avg WT	Avg TAT	Switches
FCFS	P1 → P2 → P3 → P4 → P5	4.40	9.00	4
SJF (non-preemptive)	P1 → P2 → P3 → P5 → P4	3.60	8.20	4
Priority (non-preemptive)	P1 → P2 → P3 → P5 → P4	3.60	8.20	4
Round Robin (q = 3)	P1 P2 P3 P4 P2 P5 P4 P5 P4	5.20	9.80	8

Waiting time per process	P1	P2	P3	P4	P5	Spread
FCFS	0	1	5	5	11	0-11
SJF / Priority	0	1	5	9	3	0-9
Round Robin	0	6	2	9	9	0-9

4. What the Numbers Show

- FCFS is worst on fairness though not on the averages. P5 needs 4 units of CPU and waits 11, purely because it queued behind P4. That is the convoy effect: an ATM balance check stuck behind a reconciliation job.
- SJF gives the best possible average here, 3.60, but saves nothing. At $t = 11$ it takes P5 ahead of P4, cutting P5 from 11 to 3 and pushing P4 from 5 to 9. The same total wait is shared out differently.
- Priority gives an identical schedule to SJF, which is a coincidence of these parameters, not a property of the algorithm. The priorities happen to rank the processes almost as their burst lengths do. Give P4 top priority and it becomes worse than FCFS.
- Round Robin has the worst averages and the tightest spread. Nothing waits more than 9 units, and the short urgent job P3 finishes at $t = 8$ instead of $t = 11$. It also doubles the context switches, four to eight, which the simulation does not charge for.

5. Recommendation

Round Robin with a quantum of 3, plus priority classes and aging. Average waiting time is the wrong measure here: a customer at an ATM care how long the machine takes to respond at all, not the mean across all transactions. Round Robin costs 1.60 units of average waiting and buys a bound on how long anything can wait. SJF is rejected despite the better figure because burst length is unknown when the decision has to be made, so it cannot be implemented outside a simulation. Aging removes the starvation risk the priority classes introduce.